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Online Retail Customer Value Analysis and Repurchase Behavior Prediction

Course name: INFO442 Data Science Projects

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1 Project Overview and Dataset Introduction

1.1 Project Background

This exploratory data analysis focuses on a UK cross-border gift e-commerce dataset covering transaction records from December 2010 to December 2011, spanning 13 months in total. The whole analysis is structured around four core dimensions: temporal sales trend, geographic revenue distribution, product sales performance and customer purchase behaviour. With data visualisation and multi-angle statistical analysis, implicit business patterns behind transaction data are summarised. Initial business assumptions are amended according to factual data findings, and a specific prediction modelling task is defined for subsequent research as required by coursework criteria. All charts and statistical results can be fully reproduced via the attached Jupyter Notebook code.

1.2 Dataset Basic Information

1. Observation period: 01/12/2010 – 09/12/2011, 13 calendar months;
2. Valid data scope: After excluding all return transaction records, the final dataset contains 388,804 formal sales orders;
3. Market coverage: Transactions originate from the UK and nine other European & Oceanian countries, with total overall revenue reaching £7,196,430.02;
4. Product & customer scale: The dataset includes 3,663 unique product SKUs and 4,303 distinguishable customers with complete ID information;
5. Data processing note: Raw messy data is cleaned inside the attached Python notebook to form analysis-ready data for EDA exploration.

1.3 Report Layout Compliance

This report strictly follows assignment requirements: completing univariate, bivariate and multivariate analysis, attaching detailed interpretation for all eight visualisations, updating original research hypotheses based on EDA outputs and specifying the final

modelling question. All eight qualified visual figures are embedded within the matching Jupyter Notebook file.

2 Univariate Analysis with Detailed Explanation of Eight Visualisations

2.1 Figure 1: Monthly total transaction volume trend line chart

Across 13 monitored months, the dataset accumulates 388,804 valid sales orders with an average monthly transaction volume of 29,908. Distinct seasonal fluctuation is observed: November 2011 achieves the yearly peak of 62,737 orders, boosted by pre-Christmas bulk procurement demand across European consumers. Transaction volume falls sharply in December 2011 as the core holiday shopping window closes. Monthly sales maintain mild steady growth from late 2010 to mid-2011 and accelerate from July to October before the November seasonal surge.

2.2 Figure 2: Daily transaction trend for the latest 60 operating days

The platform runs normally for 305 valid trading days without any full-day zero-sales record, with an average daily order count of 1,275 and the maximum single-day transaction hitting 3,334. Daily sales fluctuate up and down around the average value affected by weekday-weekend rotation. Only recent 60 days are selected for plotting to avoid dense date labels and simplify short-term fluctuation observation.

2.3 Figure 3: 24-hour hourly transaction distribution bar chart

Noon (12:00) becomes the daily sales peak with 70,321 aggregated orders. Regular business hours (8:00 to 18:00) generate over 98% of total platform orders (381,652 transactions). Zero purchase activity appears between midnight and 6 a.m., which proves all customers finish procurement during standard working daytime, and purchase willingness drops rapidly after 18:00.

2.4 Figure 4: Horizontal bar chart of top 10 countries sorted by cumulative revenue (Unit: Thousand GBP)

UK domestic business contributes £5,892K revenue, accounting for 84.7% of total platform turnover and forming the core business base. Netherlands, Germany, France and Ireland form the secondary market tier, whose combined revenue proportion is less than 16% of total income. The remaining five ranked nations create negligible revenue, indicating extremely uneven geographic revenue distribution and heavy market reliance on British domestic consumers.

2.5 Figure 5: Monthly aggregated revenue bar chart

Total platform revenue over the full research period equals £7,196K, with a monthly average revenue of £554K. Monthly revenue changes synchronously with monthly order quantity, and November 2011 creates the annual highest revenue (£1,014K) driven by concentrated Christmas shopping demand. Revenue rises slowly in the first half year and climbs gradually in autumn before a December decline alongside falling transaction volume.

2.6 Figure 6: Horizontal bar chart of top 10 best-selling products by sales quantity

The leading commodity JUMBO BAG RED RETROSPOT accumulates total sales of 29,868 units, far exceeding other single products. The top three SKUs together achieve total shipment of 76,136 pieces. All top-selling items belong to low-cost home storage supplies, holiday decorations and small kitchen gadgets; no high-priced large commodities enter the best-selling ranking.

2.7 Figure 7: Customer purchase frequency distribution histogram

4,303 unique registered customers have an average purchase frequency of 90.4 and a median purchase count of 41. The obvious gap between mean and median is caused by a small group of high-spending repeat clients lifting overall average value. Repeat

buyers (with ≥ 2 purchase records) occupy 98.5% of all users (4,239 people), while merely 64 customers are one-time purchasers, reflecting outstanding customer stickiness. Customers with purchase frequency over 100 are hidden to optimise histogram visual effects.

2.8 Figure 8: Weekly transaction distribution bar chart (Monday to Sunday)

Thursday records the weekly highest transaction of 78,222 orders. Orders generated from Monday to Friday account for 84.3% of total sales volume, and weekend transaction numbers shrink significantly. It can be inferred that core buyers are enterprise and office purchasers who arrange bulk purchase only on working days.

3 Bivariate Correlation Analysis

3.1 Country vs Total Revenue

There exists an unbalanced positive correlation between geographic location and generated revenue. UK domestic market is closely correlated with high turnover, while all other European countries are constrained by cross-border logistics and local consumption customs, resulting in much lower revenue scale. Statistically, UK's total revenue is around 5.5 times the aggregated revenue of all remaining nine countries, which directly proves country is a dominant factor deciding sales performance.

3.2 Weekday Attribute vs Daily Transaction Volume

Working-day property has positive correlation with daily order quantity, whereas weekend date is negatively correlated with sales volume. Order volume stays consistently high on weekdays and drops drastically on Saturdays and Sundays. Customer procurement behaviour strictly follows office working schedules rather than leisure rest time.

4 Multivariate Cross-dimensional Analysis

This section analyses the interactive relationship among month, country and product category from three dimensions:

- Seasonal month and country interaction: Nov-Dec Christmas shopping season lifts overall sales across all regions but with differentiated growth rate; UK monthly sales rise more than 60% month-on-month, while small European nations gain less than 15% sales uplift. Most seasonal consumption bonus is captured by British domestic market.
- Trading hour, product type and weekday triple interaction: Low-priced home decoration goods reach peak sales from 11:00 to 13:00 on working weekdays; the combination of working day, noon trading period and affordable small commodities becomes the core high-turnover business mode of the platform.
- Country and product preference interaction: UK customers prefer low-cost daily decorative items for bulk purchase; overseas European buyers tend to select niche featured commodities with low repurchase frequency and cannot form large-scale stable sales.

5 Initial Hypotheses, EDA Verification and Revised Hypotheses

5.1 Original Pre-analysis Hypotheses

H1: Sales revenue is distributed evenly across all cooperating countries, and European nations have limited gaps in total turnover.
H2: Customers have stronger purchase intention during free leisure weekends, leading to higher weekend order volume than weekdays. H3: High-priced premium goods contribute most platform revenue, while low-cost small products only boost sales quantity with limited profit contribution.

5.2 Hypothesis Validation via EDA Results

- H1 Rejected: UK holds 84.7% of total revenue with extreme market concentration and huge cross-country revenue disparity.
- H2 Rejected: 84.3% of all transactions happen on weekdays, and weekend consumption demand is substantially weak.
- H3 Rejected: Low-price daily necessities dominate top sales rankings and become core revenue sources; expensive commodities have poor market performance.

5.3 Final Revised Hypotheses after Data Exploration

- Platform revenue heavily relies on UK domestic market; other European countries contribute marginal revenue but have incremental growth potential via targeted refined operation.
- Core customer procurement concentrates on working daytime instead of weekend leisure hours.
- Affordable home decorative commodities dominate both sales quantity and total revenue; high-priced products cannot become main profit pillars.

5.4 Clear Modelling Statement (Mandatory coursework requirement)

Final Modelling Task: Build a supervised regression predictive model to forecast monthly total platform revenue in British Pounds.

- Independent input features: transaction month, weekday, trading hour, customer's residing country and product category;
- Target dependent variable: aggregated monthly total revenue of the whole e-commerce platform;
- Practical business purpose: Pre-estimate monthly seasonal revenue especially for pre-Christmas peak season, supporting enterprise inventory allocation, marketing budget arrangement and human resource scheduling to avoid overstock or stock-out risks.

5.5 Full Report Conclusion

Based on authentic 13-month cross-border retail transaction data, this report completes comprehensive univariate, bivariate and multivariate exploratory analysis alongside eight fully annotated visual figures as required. Original subjective hypotheses are revised with factual data evidence, and a clear regression modelling objective is confirmed. Three core business conclusions are summarised: First, the UK domestic market is the core revenue foundation, and continental European markets are potential incremental growth space. Second, weekday noon and November–December holiday period are two annual core sales peaks. Third, low-cost home accessories are the dominant driver of both sales volume and platform revenue with outstanding customer repurchase loyalty. All analysis results are fully reproducible by running the attached Jupyter Notebook code.